Module 8: Introduction



Syllabus (https://northeastern.instructure.com/courses/257815/assignments/sy

Modules (https://northeastern.instructure.com/courses/257815/modules)

Resources (https://northeastern.instructure.com/courses/257815/pages/course-

Assignments (https://northeastern.instructure.com/courses/257815/assignment

Module Overview

This module examines the intersection of AI and robotics, concentrating on how intelligent systems can perceive and interact with the physical world. Students will learn about robot kinematics and dynamics, sensor fusion, simultaneous localization and mapping (SLAM), path planning, and control systems. The module covers both industrial robotics applications and emerging areas like social robotics, swarm robotics, and human-robot interaction.

Learning Objectives

By the end of this module, you should be able to learn about:

- The fundamental concepts of robotics and embodied intelligence
- How robots sense, think, and act in the real world
- The evolution from traditional robotics to AI-powered physical systems
- Real-world applications transforming our daily lives

Task List

- Participate in the weekly discussion board
- Solve and submit the weekly assignment
- Additional reading:
  - maizi5469. (2024, November 30). *2.1 Sensors and Actuators in Robotics*. Medium. **https://medium.com/@maizi5469/2-1-sensors-and-actuators-in-robotics-968a67aee969 (https://medium.com/@maizi5469/2-1-sensors-and-actuators-in-robotics-968a67aee969)**
  - Communications of the ACM. (2024, May 21). *A Brief History of Embodied Artificial Intelligence, and its Future Outlook*. **https://cacm.acm.org/blogcacm/a-brief-history-of-embodied-artificial-intelligence-and-its-future-outlook/ (https://cacm.acm.org/blogcacm/a-brief-history-of-embodied-artificial-intelligence-and-its-future-outlook/)**
  - Unknown (2010). Actuators and Sensors. In *[Book Title]* (pp. [Page Range if known]). Springer. **https://link.springer.com/chapter/10.1007/978-1-84628-642-1_5 (https://link.springer.com/chapter/10.1007/978-1-84628-642-1_5)**
  - Wang, Z., Huang, Q., Wang, Y., Zhang, Z., Yuan, Q., Xia, S., ... & Ouyang, W. (n.d.). *Aligning Cyber Space with Physical World: A Comprehensive Survey on Embodied AI*. arXiv. Retrieved July 20, 2025, from **https://arxiv.org/html/2407.06886v1 (https://arxiv.org/html/2407.06886v1)**
  - Acrome Robotics. (2024, October 18). *Control Systems: What is PID Control?* **https://acrome.net/post/control-systems-what-is-pid-control (https://acrome.net/post/control-systems-what-is-pid-control)**
  - Correll, N., Hayes, B., Heckman, C., & Roncone, A. (2022). *Introduction to Autonomous Robots: Mechanisms, Sensors, Actuators, and Algorithms*. MIT Press.
  - Weijie Zhao, Ye Yuan, Development of intelligent robots in the wave of embodied intelligence, *National Science Review*, Volume 12, Issue 7, July 2025, nwaf159, **https://doi.org/10.1093/nsr/nwaf159 (https://doi.org/10.1093/nsr/nwaf159)**

- Li, J.; Yang, S. X. Digital twins to embodied artificial intelligence: review and perspective. *Intell. Robot.* **2025**, *5*(1), 202-27. http://dx.doi.org/10.20517/ir.2025.11
- Huaping Liu, Di Guo, and Angelo Cangelosi. 2025. Embodied Intelligence: A Synergy of Morphology, Action, Perception and Learning. ACM Comput. Surv. 57, 7, Article 186 (July 2025), 36 pages. **https://doi.org/10.1145/3717059 (https://doi.org/10.1145/3717059)**
- Mon-Williams, R., Li, G., Long, R. *et al.* Embodied large language models enable robots to complete complex tasks in unpredictable environments. *Nat Mach Intell* **7**, 592–601 (2025). **https://doi.org/10.1038/s42256-025-01005-x (https://doi.org/10.1038/s42256-025-01005-x)**
- STEMRobotics. (n.d.). *Exercise: Using a PID Controller*. Retrieved July 20, 2025, from **https://stemrobotics.cs.pdx.edu/node/7268.html (https://stemrobotics.cs.pdx.edu/node/7268.html)**
- Zhao, Z., Wu, Q., Wang, J., Zhang, B., Zhong, C., & Zhilenkov, A. A. (2024). Exploring Embodied Intelligence in Soft Robotics: A Review. *Biomimetics (Basel, Switzerland)*, *9*(4), 248. **https://doi.org/10.3390/biomimetics9040248 (https://doi.org/10.3390/biomimetics9040248)**
- Unknown (2024, January 10). *How to Use Sensors and Actuators for Robotics*. LinkedIn. **https://www.linkedin.com/advice/0/how-do-you-use-different-sensors-actuators-skills-robotics (https://www.linkedin.com/advice/0/how-do-you-use-different-sensors-actuators-skills-robotics)**
- FIRST Robotics Competition. (2025, June 2). *Introduction to PID*. WPILib Documentation. **https://docs.wpilib.org/en/stable/docs/software/advanced-controls/introduction/introduction-to-pid.html (https://docs.wpilib.org/en/stable/docs/software/advanced-controls/introduction/introduction-to-pid.html)**
- Autonomous Robots Lab. (n.d.). *PID Control*. Retrieved July 20, 2025, from **https://www.autonomousrobotslab.com/pid-control.html (https://www.autonomousrobotslab.com/pid-control.html)**
- Wikipedia. (n.d.). *Proportional–integral–derivative controller*. Retrieved July 20, 2025, from **https://en.wikipedia.org/wiki/Proportional%E2%80%93integral%E2%80%93derivative_controller (https://en.wikipedia.org/wiki/Proportional%E2%80%93integral%E2%80%93derivative_controller)** *(Note: As Wikipedia is a living document, a retrieval*

date is important. "Updated 5 days ago" indicates continuous updates, so n.d. for publication date is appropriate.)

- *rrwiyatn. (2020, June 15). Robotik - Ep.4: Intro to Control and Proportional-Integral-Derivative (PID) Controller. rey's blog.* ***https://rrwiyatn.github.io/blog/robotik/2020/06/15/pid.html*** ***(https://rrwiyatn.github.io/blog/robotik/2020/06/15/pid.html)***
- Chidanand Hegde, Jiangtao Su, Joel Ming Rui Tan, Ke He, Xiaodong Chen, and Shlomo Magdassi
  ACS Nano **2023** *17* (16), 15277-15307, *DOI: 10.1021/acsnano.3c04089*
- *FutureLearn. (2022, October 25). Sensors and Actuators.* ***https://www.futurelearn.com/info/courses/begin-robotics/0/steps/2860*** ***(https://www.futurelearn.com/info/courses/begin-robotics/0/steps/2860)***
- *jaems33. (2018, June 1). Understanding Robot Motion: PID Control. Medium.* ***https://medium.com/@jaems33/understanding-robot-motion-pid-control-8931899c31df*** ***(https://medium.com/@jaems33/understanding-robot-motion-pid-control-8931899c31df)***
- *Manuel Beschi, Riccardo Adamini, Alberto Marini, Antonio Visioli (2015), Using of the Robotic Operating System for PID control education, IFAC-PapersOnLine, Volume 48, Issue 29, 87-92, https://www.sciencedirect.com/science/article/pii/S2405896315024763*
- *Valin. (n.d.). Using PID for Motion Control, Robotics. Retrieved July 20, 2025, from* ***https://www.valin.com/resources/articles/using-pid-motion-control-robotics*** ***(https://www.valin.com/resources/articles/using-pid-motion-control-robotics)***
- *Encord. (2025, March 26). What is Embodied AI? A Guide to AI in Robotics.* ***https://encord.com/blog/embodied-ai/*** ***(https://encord.com/blog/embodied-ai/)***
- *Wang, J., Herath, D. (2022). What Makes Robots? Sensors, Actuators, and Algorithms. In: Herath, D., St-Onge, D. (eds) Foundations of Robotics. Springer, Singapore. https://doi.org/10.1007/978-981-19-1983-1_7*